AI SL — Specific Topics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • **Target: Solutions to trigonometric equations**

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(5)$.

Final Answer: 23. Substitute $x = 5$: $f(5) = 4(5) + 3 = 23$.

2 [Foundational] • **Target: Solutions to trigonometric equations**

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

3 [Foundational] • **Target: Problem solving with trigonometric functions**

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(2)$.

Final Answer: 14. Substitute $x = 2$: $f(2) = 5(2) + 4 = 14$.

4 [Foundational] • **Target: Problem solving with trigonometric functions**

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

5 [Foundational] • **Target: Problem solving with trigonometric functions**

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

6 [Foundational] • **Target: Problem solving with trigonometric functions**

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(3)$.

Final Answer: 19. Substitute $x = 3$: $f(3) = 5(3) + 4 = 19$.

7 [Foundational] • **Target: Problem solving with trigonometric functions**

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Problem solving with trigonometric functions

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(4)$.

Final Answer: 24. Substitute $x = 4$: $f(4) = 5(4) + 4 = 24$.

9 [Intermediate] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

10 [Intermediate] • Target: Problem solving with trigonometric functions

For the model $f(x) = 5x + 4$ in problem solving with trigonometric functions, evaluate $f(5)$.

Final Answer: 29. Substitute $x = 5$: $f(5) = 5(5) + 4 = 29$.

11 [Intermediate] • Target: Problem solving with trigonometric functions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

12 [Intermediate] • Target: Double angle identities

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 86°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 45^\circ) = 86^\circ$.

13 [Intermediate] • Target: Double angle identities

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

14 [Intermediate] • Target: Double angle identities

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 7. Substituting $x = 1$ into $y = 2x + 5$ gives $y = 2(1) + 5 = 7$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 79°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 52^\circ) = 79^\circ$.

16 [Challenging] • Target: Double angle identities • Cross-Lesson: 1.01 Number & Algebra

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 24 cm. Multiplying the original length by the scale factor gives $8 \times 3 = 24$ cm.

17 [Challenging] • Target: Double angle identities • Cross-Lesson: 2.01 Functions

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 9. Substituting $x = 2$ into $y = 2x + 5$ gives $y = 2(2) + 5 = 9$.

18 [Challenging] • Target: Double angle identities • Cross-Lesson: 3.01 Geometry & Trigonometry

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 72°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 59^\circ) = 72^\circ$.

19 [Challenging] • Target: Double angle identities • Cross-Lesson: 4.01 Statistics & Probability

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 32 cm. Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

20 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 11. Substituting $x = 3$ into $y = 2x + 5$ gives $y = 2(3) + 5 = 11$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(2)$.

a. 12 ✓

b. 14

c. 4

d. 16

Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

22 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

23 [Foundational] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(5)$.

a. 18 ✓

b. 20

c. 10

d. 22

Substitute $x = 5$: $f(5) = 2(5) + 8 = 18$.

24 [Foundational] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(2)$.

a. 15 ✓

b. 18

c. 6

d. 19

Substitute $x = 2$: $f(2) = 3(2) + 9 = 15$.

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25 [Foundational] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(5)$.

a. 24 ✓

b. 27

c. 15

d. 28

Substitute $x = 5$: $f(5) = 3(5) + 9 = 24$.

27 [Foundational] • Target: Solutions to trigonometric equations

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(2)$.

a. 11 ✓

b. 15

c. 8

d. 15

Substitute $x = 2$: $f(2) = 4(2) + 3 = 11$.

28 [Foundational] • Target: Solutions to trigonometric equations

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

A data set recorded for amplitude, vertical shifts and dilations of trigonometric graphs consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

31 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

For the model $f(x) = 2x + 8$ in amplitude, vertical shifts and dilations of trigonometric graphs, evaluate $f(4)$.

a. 16 ✓

b. 18

c. 8

d. 20

Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

32 [Intermediate] • Target: Amplitude, vertical shifts and dilations of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

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33 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

34 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

For the model $f(x) = 3x + 9$ in period and phase shifts of trigonometric graphs, evaluate $f(3)$.

a. 18 ✓

b. 21

c. 9

d. 22

Substitute $x = 3$: $f(3) = 3(3) + 9 = 18$.

35 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

36 [Intermediate] • Target: Period and phase shifts of trigonometric graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

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37 [Intermediate] • Target: Solutions to trigonometric equations

A data set recorded for solutions to trigonometric equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5 **b.** 21.5

c. 22.5 ✓ **d.** 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solutions to trigonometric equations • Cross-Lesson: 3.01 Geometry & Trigonometry

For the model $f(x) = 4x + 3$ in solutions to trigonometric equations, evaluate $f(4)$.

a. 19 ✓ **b.** 23

c. 16 **d.** 23

Substitute $x = 4$: $f(4) = 4(4) + 3 = 19$.

39 [Challenging] • Target: Problem solving with trigonometric functions • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for problem solving with trigonometric functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5 **b.** 22.5

c. 23.5 ✓ **d.** 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

40 [Challenging] • Target: Double angle identities • Cross-Lesson: 5.01 Calculus

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm **b.** 16 cm ✓

c. 24 cm **d.** 32 cm

Multiplying the original length by the scale factor gives $8 \times 2 = 16$ cm.

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41 [Challenging] • Target: Double angle identities • Cross-Lesson: 1.01 Number & Algebra

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 79° ✓

b. 89°

c. 71°

d. 131°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 52^\circ) = 79^\circ$.

42 [Challenging] • Target: Double angle identities • Cross-Lesson: 2.01 Functions

In ΔABC studying double angle identities, $m\angle A = 49^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 72° ✓

b. 82°

c. 64°

d. 131°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (49^\circ + 59^\circ) = 72^\circ$.

43 [Challenging] • Target: Double angle identities • Cross-Lesson: 3.01 Geometry & Trigonometry

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm ✓

c. 40 cm

d. 128 cm

Multiplying the original length by the scale factor gives $8 \times 4 = 32$ cm.

44 [Challenging] • Target: Double angle identities • Cross-Lesson: 4.01 Statistics & Probability

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 11

b. 15

c. 13 ✓

d. 15

Substituting $x = 4$ into $y = 2x + 5$ gives $y = 2(4) + 5 = 13$.

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45 [Challenging] • Target: Linear regression line • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 5x + 5$ in linear regression line, evaluate $f(3)$.

a. 20 ✓

b. 25

c. 15

d. 24

Substitute $x = 3$: $f(3) = 5(3) + 5 = 20$.

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